



Enterprise AI Algo Trading Platform

Complete Institutional Quantitative Trading System & Operations Guide

Version 1.0.0 • Prepared for Raman Singh

Platform Overview: Institutional-grade quantitative algorithmic trading software designed for automated paper and live market execution across 132 quantitative strategies with dynamic risk management, circuit breakers, and automated execution.

1. User Roles & Architecture Overview

Role	Access URL	Primary Objective
 Admin	<code>http://localhost:5173/admin/dashboard</code>	System Risk, Broker Connections, Live Gate Approval, Master Kill-Switch, Audit Trail, Trade Reconciliation
 Trader	<code>http://localhost:5173/dashboard</code>	132 Strategies Autopilot, Paper/Live Trading, Signal Approvals, Live Orders, Portfolio & Smart Journal

2. Admin Dashboard & Enterprise Capabilities

- 🛡️ **Dynamic Risk Dashboard** (`/admin/risk`): Real-time Exposure meter, Daily Loss Limit breach alerts, Order Velocity throttler, and Paper Account Reset (₹10,000 to ₹1,00,000 presets).
- 🔴 **Master Kill Switch** (`/admin/kill-switch`): Emergency 1-click circuit breaker to immediately halt all active strategies and cancel open limit orders.
- 🔒 **Live Trading Gate** (`/admin/live-gate`): Institutional safety lock requiring paper trading performance certification before real money execution is permitted.
- 🔗 **Multi-Broker Manager** (`/admin/brokers`): Zerodha Kite, Angel One, Upstox, Dhan, Fyers, and Simulated Paper Sandbox broker integrations.
- ⚖️ **Reconciliation** (`/admin/reconciliation`): Real-time exchange vs. local database trade state mismatch detection and synchronization.
- 📄 **Audit Trail** (`/admin/audit`): Immutable event logging for compliance and governance tracking.

3. Trader Capabilities & Autopilot Strategy Engine

- ⚡ **132 Quant Strategies Autopilot** (`/strategies`): Rule-Based, ML Classification, FinBERT NLP Sentiment, Pairs Arbitrage, Mean-Reversion, and Momentum Scalping.
- 🔧 **Dual-Action Individual Card Controls:**
 - When Strategy is **Stopped**: Displays **▶ Resume Strategy** (Green). Clicking it immediately turns the strategy **● Active** and renders **▢ Stop Strategy** (Red).
 - When Strategy is **Active**: Displays **▢ Stop Strategy** (Red). Clicking it immediately stops the strategy and renders **▶ Resume Strategy** (Green).

- 🚀 **Global Batch Actions:**
 - 🚀 **Deploy All to Paper** : Deploys and activates all 132 strategies simultaneously in paper sandbox mode.
 - 🛑 **Stop All** : Safely halts all 132 strategies in 1 click.
- 💰 **Dynamic 1% Risk Position Sizing:**

$$\text{Quantity} = \min(\text{floor}((\text{Cash Balance} \times 0.01) / (\text{Entry Price} - \text{Stop Loss})), \text{floor}(\text{Max Capital} / \text{Entry Price}))$$

Ensures maximum capital loss per trade is strictly capped at 1% (₹100 on ₹10,000 capital).
- 📁 **Portfolio & Orders (/orders , /portfolio)**: Real-time P&L tracking, trailing stop-losses, and manual square-off controls.
- 📖 **Smart Journal (/journal)**: Automated trade logging with entry/exit logic and performance analytics.

4. Step-by-Step Daily Trading Workflow (Monday to Friday)

- 🌅 **Pre-Market (09:00 AM - 09:15 AM):**
 - Open `http://localhost:5173/strategies`
 - Verify System Health status is Green.
 - Click 🚀 **Deploy All to Paper** (or resume selected top strategies).
- 🔔 **Live Market Hours (09:15 AM - 03:30 PM):**
 - Strategies evaluate live ticks and generate quantitative signals.
 - In **Auto-Pilot Mode**: System auto-places orders, tracks ATR Trailing SL, and exits at target.
 - In **Advisory Mode**: System requests trader 1-click confirmation before executing orders.
- 🌃 **Post-Market (03:30 PM - 04:00 PM):**
 - Review Daily Net P&L and Win Rate in `/dashboard` .
 - Click 🛑 **Stop All** to park strategies in standby mode.
 - Review automatically logged trades in `/journal` .

5. Paper Trading vs Live Broker Trading

Parameter	📘 Paper Trading (Virtual Sandbox)	🔴 Live Broker Trading
Financial Risk	₹0.00 (Zero Risk)	Real Capital at Risk
Account Balance	Virtual ₹10,000 (Resettable anytime)	Real Demat Account Balance
Execution Engine	Simulated exchange tick execution	Direct broker API order routing
Primary Goal	Strategy validation, win-rate & edge testing	Real profit generation on proven setups

Recommended Practice: Run 2 full weeks of paper testing using the virtual ₹10,000 budget with 1% risk management before deploying any live capital.

